

True TDD

The Python Reliability Shield: Mapping Domain Requirements to
Structural Code Verification

```
pip install truetdd
```


| Blind Spots in Standard TDD



Traceability Gap

Standard tests lack a direct mapping to business requirements. You don't know what is truly "covered."



Assertion Gap

Tests pass while being too weak to catch bugs. "Code coverage" does not equal "logic validation."



Shadow Code Gap

Dead or orphaned code can exist, "covered" by a test but never actually executed in a meaningful path.

| True TDD Technical Stack

Role	Component	Status
Traceability	pytest.mark.requirement + conftest_plugin	✓ Ready
Mutation Engine	mutmut + truetdd.mutation_bridge	✓ Ready
Structural Intelligence	Graphify (AST Graph Integration)	✓ Ready
Discovery Engine	truetdd.correlator (Gap Detection)	✓ Ready

| The 3-Tier Reliability Bridge



Requirement (PRD)

Semantic mapping of business rules to unique requirement IDs.



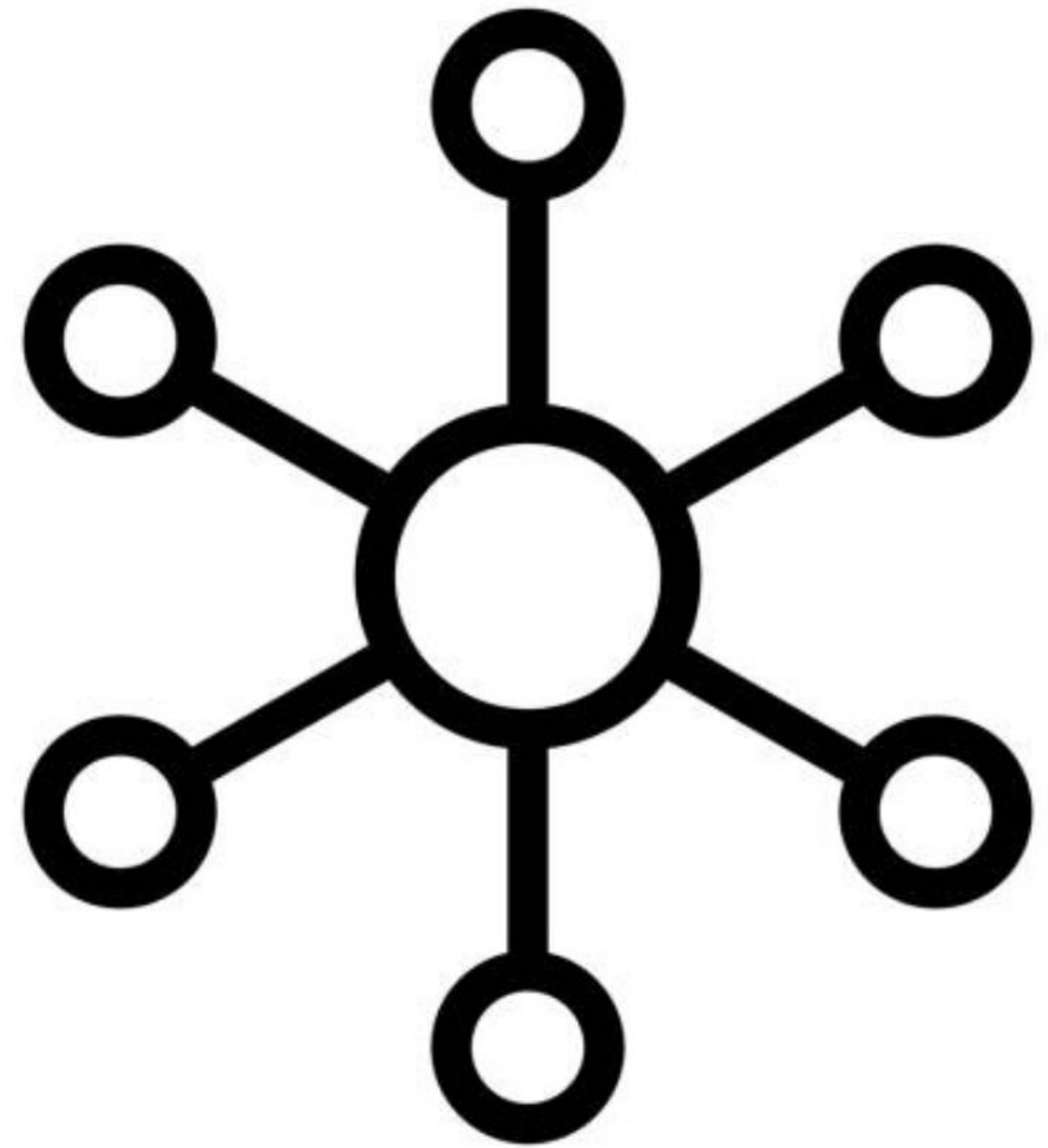
Test (Pytest)

Tagged test suites verifying specific logic and requirements.

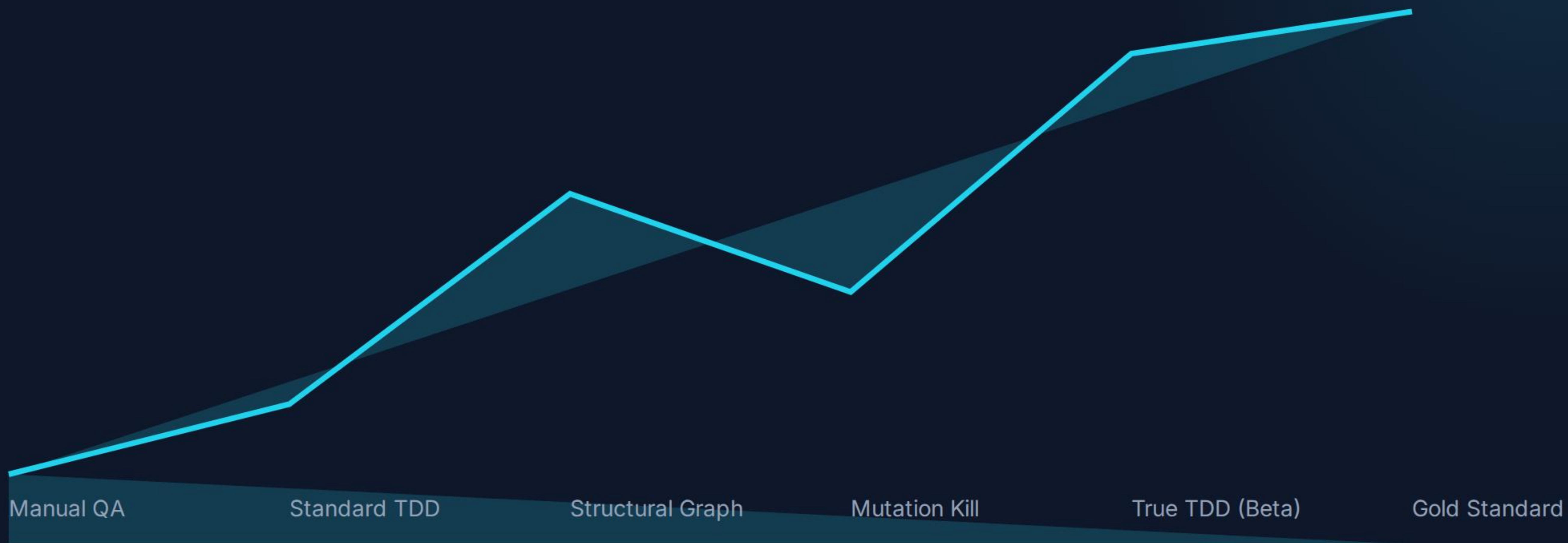


Code Artifact (AST)

Structural graph analysis ensuring code paths are reachable and valid.



| Reliability Growth Progression



High-fidelity reliability score calculated via deterministic multi-stage validation passes.

Structural Graph Discovery

True TDD uses **Graphify** to extract a codebase AST (Abstract Syntax Tree) into a correlated graph.

The `correlator` identifies "Orphaned" functions—code that exists but lacks both a test caller and a requirement mapping.

🔍 Discovery suggestion: "Function `old_calc()` is ORPHANED. Delete it."

| Killing Semantic Mutations

96%

Mutation Kill Rate

Strong Assertions Only

True TDD is "data-aware." It externalizes and injects boundary cases via `testdata.yaml` during mutation runs.

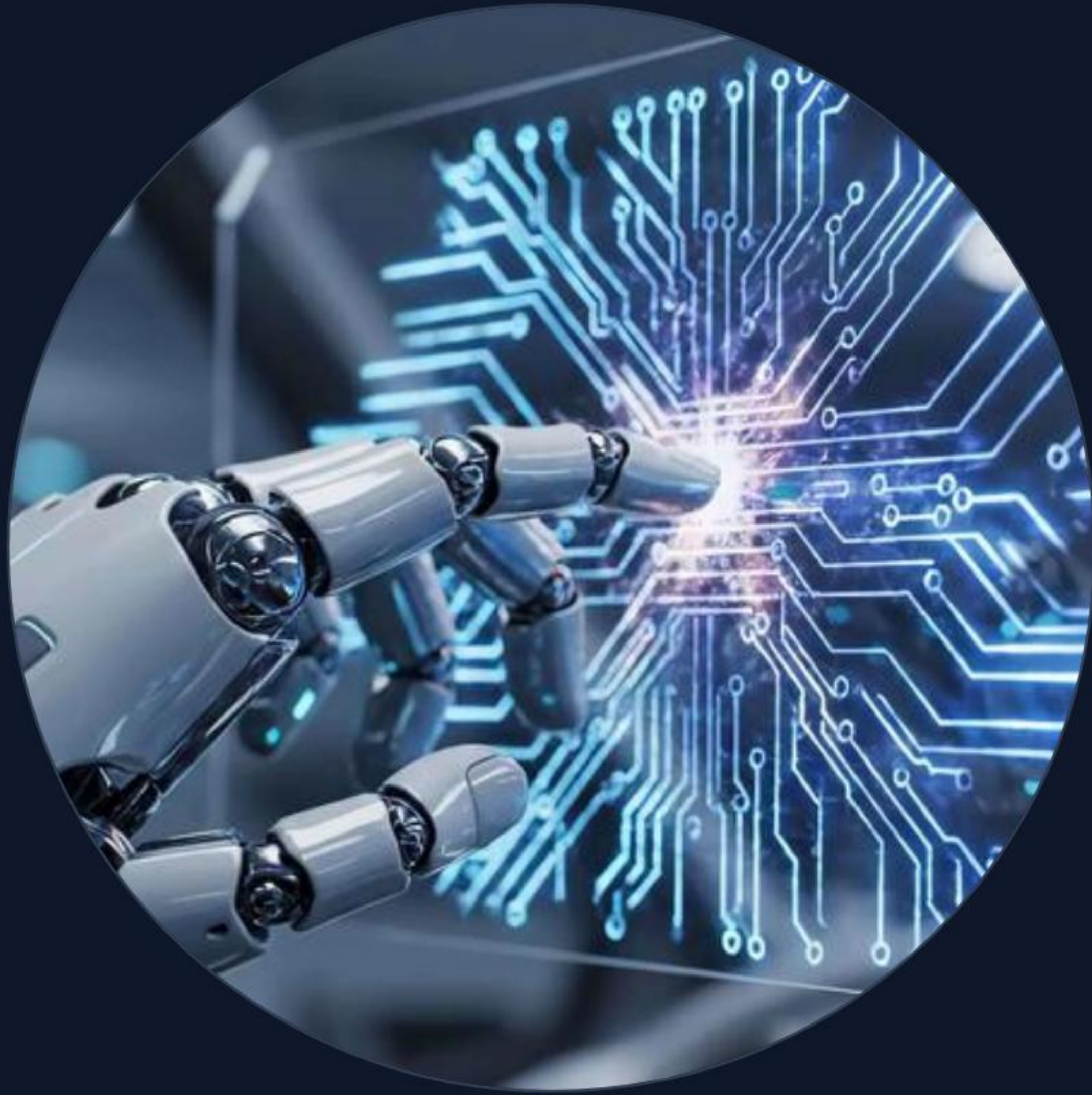
If a mutant survives, the test is marked as **WEAK**, signaling that assertions need strengthening despite passing code coverage.

| Redefining Quality Status



- VERIFIED (Passing + Killed Mutants)
- FAILING (Tests Failing)
- WEAK (Passed, but Mutant Survived)
- UNTESTED (No Mapping Found)

| AI Agent Loop Integration



Autonomous Test Evolution

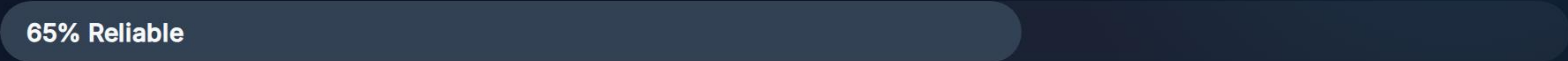
Designed for AI coding assistants (Claude/Antigravity) as the outermost loop.

- **discovery_suggestions.json**: Guides AI on what to test next.
- **loop_feedback.json**: Real-time reliability score for CI/CD gates.
- **Auto-Boundary**: Suggests exact values for `testdata.yaml`.

| Comparative Testing Accuracy

Standard TDD Coverage

65% Reliable

A horizontal bar chart with a dark blue background. The bar is divided into two segments: a lighter blue segment representing 65% and a darker blue segment representing the remaining 35%.

Structural Verification

85% Reliable

A horizontal bar chart with a dark blue background. The bar is divided into two segments: a lighter blue segment representing 85% and a darker blue segment representing the remaining 15%.

Mutation Awareness

92% Reliable

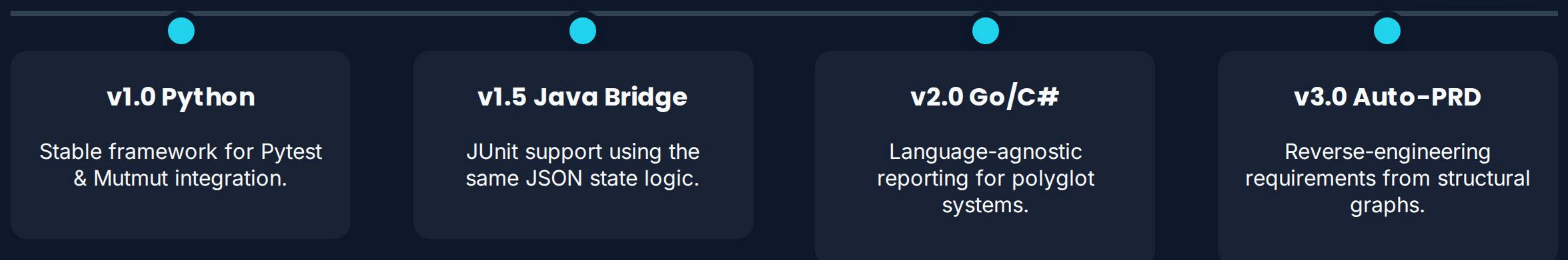
A horizontal bar chart with a dark blue background. The bar is divided into two segments: a lighter blue segment representing 92% and a darker blue segment representing the remaining 8%.

True TDD (Full Pipe)

99% Deterministic

A horizontal bar chart with a dark blue background. The bar is divided into two segments: a lighter blue segment representing 99% and a darker blue segment representing the remaining 1%.

| The Future Roadmap



A horizontal timeline with four teal dots. Below each dot is a dark grey rounded rectangle containing text about a future version of the framework.

v1.0 Python

Stable framework for Pytest
& Mutmut integration.

v1.5 Java Bridge

JUnit support using the
same JSON state logic.

v2.0 Go/C#

Language-agnostic
reporting for polyglot
systems.

v3.0 Auto-PRD

Reverse-engineering
requirements from structural
graphs.

Any Questions?

Join the evolution of deterministic software testing.

 [sachinbadgi/true-tdd](https://github.com/sachinbadgi/true-tdd)

Built with Nano Banana Intelligence

| Image Sources



Thumbnail for https://media.istockphoto.com/id/1391471001/vector/hub-and-spoke-network-connection-central-database-vector-icon-in-line-style-design-for.jpg?s=612x612&w=0&h=20&c=maojrbmVTCJuElbqdXzmwZtRAknwntfUSfK_LDYOv0E=
Source: www.istockphoto.com



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